Sham Satish Thakare

Independent Computer Science Researcher
Data Engineer / Machine Learning Specialist at Spectra Corporate Service
Pune, Maharashtra, India • Email: shamthakare3000@gmail.com • Web: https://shamddd.github.io/shamthakare.github.io • GitHub: https://github.com/shamddd

Education

APCOER Pune — Pune, India
B.Tech. in Artificial Intelligence & Data Science | *Jan 2020 – Jun 2024*

- Cumulative GPA: **8.70 / 10.00**
- Core Focus: Machine Learning, Probability & Statistics, Data Structures & Algorithms, Distributed Systems.

Research / Professional Experience

Spectra Corporate Service — Pune, India
Data Engineer / Machine Learning Specialist | *Jun 2024 – Present*

- Engineered high-throughput parallel data processing systems and ML data pipelines using C++, Python, and SQL, optimizing memory layout and execution locality.
- Designed multithreaded cloud workflow automation and telemetry data processing pipelines enforcing concurrency invariants and data integrity.
- Implemented infrastructure performance benchmarking and automated execution suites across compute clusters to measure scale throughput.

APCOER Pune — Pune, India
Research Intern | *Jun 2023 – Dec 2024*

- Conducted independent research on machine learning algorithms and statistical data evaluation methods.
- Provided instructional and teaching support for undergraduate coursework in *Data Structures, Algorithms, and Foundations of Machine Learning*.
- Developed open-source benchmark scripts and automated test suites for algorithmic problem solving.

Manuscripts & Publications

Submitted / Under Review

When Confidence Proxies Confound Reasoning Complexity: Pitfalls of Uncertainty-Weighted Credit Assignment in Language Model Reinforcement Learning
Sham Satish Thakare
Submitted to IEEE Transactions on Artificial Intelligence (IEEE TAI), 2026.
Falsifies token-predictive entropy as an uncertainty proxy in LM RLVR. Demonstrates predictive entropy correlates strongly with sequence length ($r = +0.486$), causing uncertainty-weighted GRPO to misallocate credit on valid multi-step derivations.

recovery_eval: State-Matched and Provenance-Aware Evaluation of Recovery Behavior in Language-Model Reasoning
Sham Satish Thakare
Submitted to 11th IEEE Special Session on Machine Learning on Big Data (IEEE BigData 2026 / MLBD, Submission ID: BigD497), 2026.
Formulates state-contrast continuation evaluation ($D_{\text{recovery}} = -0.1100$) measuring error recovery under frozen structural covariates across reasoning model rollouts.

StateShift: State-Matched Evaluation of Error Recovery in Neural Reasoning
Sham Satish Thakare
Submitted to Elsevier Artificial Intelligence (AIJ, Manuscript ID: ARTINT-D-26-01491), 2026.
Evaluates neural reasoning continuation state matching under verifier perturbations and structural covariate controls.

Working Papers

Amortized Intervention Frontiers for Language-Model Reasoning: When Does Training Beat Search?
Sham Satish Thakare | *Working Paper, 2026 (Target under consideration: TMLR)*

Predicting Reinforcement-Learning Plasticity of Intermediate Language-Model Checkpoints
Sham Satish Thakare | *Working Paper, 2026 (Target under consideration: JMLR)*

EnclaveShield: Zero-Knowledge Memory Attestation and Side-Channel Mitigation for Hardware Enclaves
Sham Satish Thakare | *Working Paper, 2026 (Target under consideration: IEEE TDSC)*

AdaptiveReplica: Dynamic Quorum Adaptation and Failure-Aware Replica Selection in Distributed Consensus
Sham Satish Thakare | *Working Paper, 2026 (Target under consideration: IEEE TPDS)*

TraceMind: Graph-Constrained Causal Reasoning for Root-Cause Localization in Microservice Systems
Sham Satish Thakare | *Working Paper, 2026 (Target under consideration: IEEE TCC)*

Research Software & Artifacts

- **ear_grpo_reasoning**: LM RL credit assignment probing suite exposing length confounding in token predictive entropy. (`https://github.com/shamddd/ear_grpo_reasoning`)
- **recovery_eval**: State-matched continuation evaluation protocol measuring error recovery under frozen structural covariates. (`https://github.com/shamddd/recovery_eval`)

• **quorumshift**: Failure-domain aware dynamic quorum adaptation in C++20 Raft consensus clusters.
(`https://github.com/shamddd/quorumshift`)
• **enclaveshield**: Zero-knowledge memory attestation proofs and adaptive Path ORAM tree rebalancing.
(`https://github.com/shamddd/enclaveshield`)

Selected Technical Skills

• **Research & ML**: Reinforcement Learning (GRPO/PPO), Foundation Model Evaluation, PyTorch, Model Probing, Statistical Hypothesis Testing.
• **Systems Research**: Distributed Consensus (Raft), Confidential Computing (Path ORAM/ZK Attestation), C++20 Multithreading, Concurrency Controls.
• **Professional Engineering**: High-Throughput Data Pipelines, Linux Systems Administration, SQL Optimization, Docker, Cloud Telemetry.
• **Programming Languages**: C++, Python, SQL, Bash.